

MOHAMMADALI AMERI FARD NASRAND

Curriculum Vitae

PERSONAL AND CONTACT INFORMATION

First Name: **MohammadAli**
Email: **amerifar@ualberta.ca**

Last Name: **Ameri Fard Nasrand**
Linkedin: www.linkedin.com/in/mohammad-ali-ameri

SUMMARY

Ph.D. candidate in Structural Engineering at the University of Alberta with expertise in risk and reliability analysis, numerical modeling, and pipeline integrity. Experienced in risk-based inspection and maintenance planning, integrating Bayesian decision analysis, reinforcement learning, and probabilistic methods for pipeline integrity management. Strong background in structural health monitoring, finite element analysis (FEA), and predictive modeling for pipeline degradation under stress corrosion cracking (SCC) and fatigue growth. Proven ability to collaborate on industry-focused research, publish in high-impact journals, and develop AI-driven solutions for pipeline integrity management.

EDUCATION

University of Alberta, Canada PhD Student, Structural Engineering Department of Civil Engineering	<i>Sep 2021 - Oct 2026</i> Overall GPA: 3.9/4
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Sharif University of Technology, Iran M.Sc., Structural Engineering Department of Civil Engineering	<i>Sep 2018 - Jan 2021</i> GPA: 17.86/20 (4/4) (Top 15%)
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Sharif University of Technology, Iran B.Sc., Civil Engineering Department of Civil Engineering	<i>Sep 2014 - Sep 2018</i> GPA: 17.23/20 (3.75/4) (Top 20%, Honor Student)
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WORK EXPERIENCE

Ph.D. Researcher at the University of Alberta	Sep 2021 – June 2026
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• **Ph.D. Project:** *Risk-Based Inspection and Maintenance Planning for Pipeline Integrity Management against Cracks* (Supervisor: Prof. Yong Li)

- 1) Developed decision-making frameworks in Python, integrating Bayesian analysis, reinforcement learning, and reliability-based optimization to enhance pipeline integrity management.
- 2) Performed probabilistic failure analysis using Monte Carlo simulation and FORM/SORM methods to assess cracked pipelines reliability under stress corrosion cracking (SCC) and fatigue.
- 3) Built Python-based predictive models for pipeline maintenance scheduling, combining statistical analysis with AI-driven optimization.
- 4) Optimized inspection intervals and repair thresholds using Value of Information (VoI) and cost-benefit analysis under uncertainty.
- 5) Working under supervision of Prof. Samer Adeeb to parallelize Monte Carlo sampling for structural reliability purposes in MecSimCalc platform.¹

¹<https://docs.mecsimcalc.com/advanced-applications/mp-monte-carlo>

- 6) **Lab2Market Program:** Participated in entrepreneurship training focused on translating academic research into practice through customer discovery, market analysis, and business model development for pipeline integrity management software.

M.Sc. Researcher at Sharif University

Sep 2018 – Aug 2021

M.Sc. Project: *Probabilistic Structural Health Monitoring for Rapid Damage Detection after Earthquakes* (Supervisor: Prof. Mahsuli)

- 1) Designed a Bayesian-based framework in MATLAB for probabilistic system identification and rapid post-earthquake damage detection.
- 2) Integrated sensor and mobile data to support real-time decision-making in intelligent emergency response systems.
- 3) Modeled and analyzed different concrete and steel structures in SAP2000 and ETABS to evaluate damage potential under seismic excitations.
- 4) Contributed to developing probabilistic models that improved regional emergency resource allocation and resilience planning.

Internship – Municipality of Tehran

Jul 2018 – Sep 2018

- 1) Gained practical experience in site supervision, quantity estimation, labor management, and construction safety compliance.
- 2) Supported project design and documentation processes for municipal infrastructure development.
- 3) Checked and validated structural designs in ETABS and AutoCAD.

TECHNICAL STRENGTHS

Programming	Professional in Python, Matlab, C++, Power BI
Pipeline Risk Management	API 580/581, API 579, CSA Z662, RBI Planning
Structural Design	Abaqus, FEM, SAP2000, ETABS, AutoCAD, OpenSees
Structural Health Monitoring	Kalman Filter, Signal Processing, Bayesian Methods

TEACHING ASSISTANTSHIPS

Statics/Dynamics/Math 102 at ESSC, UofA	Sep 2022 - Present
Co-up Marker, UofA	April 2025
Numerical Methods for Engineers	Jan-Apr 2022
Subsection Resilience of Infrastructures	Feb-Jun 2020
Introduction to Earthquake Engineering	Feb-Jun 2020
Concrete Technology Laboratory	Sep-Dec 2017

PUBLICATIONS

Journal Papers:

- 1) M.A Ameri Fard Nasrand, M. Mahsuli, S. F. Ghahari, E. Taciroglu, "Bayesian Model Selection Considering Model Complexity Using the Extended Kalman Filter", *Earthquake Engineering and Structural Dynamics*, 2023. doi.org/10.1002/eqe.3915
- 2) M.A Ameri Fard Nasrand, Y. Li, "Risk-based Decision-making for Immediate Re-inspection of Pipe Cracks using Value of Information Analysis", *ASCE Journal of Pipeline Systems Engineering and Practice*, 2026. (in press)
- 3) M.A Ameri Fard Nasrand, Y. Li, "Dynamic Inspection and Maintenance Planning for Cracked Pipes using Reinforcement Learning with Risk-informed Rewards", *Reliability Engineering and System Safety*, 2026. (in press)
- 4) M.A Ameri Fard Nasrand, Y. Li, "Risk-based Scheduling for Multiple Delayed Inspections of Less Severe Cracks in Pipes considering Uncertain Crack Growth", *Journal of Pipeline Systems Engineering and Practice*, 2026. (To be submitted)
- 5) M.A Ameri Fard Nasrand, Y. Li, "Risk-Based Inspection Planning for Pipeline System containing Multiple Joints considering Uncertain Crack Birth and Growth", *Reliability Engineering and System Safety*, 2026. (To be submitted)
- 6) M.A Ameri Fard Nasrand, Y. Li, "Delayed Inspection Planning with Optimized Reliability-Based Repair Criteria for Pipelines with Single and Population of Growing Cracks", *Structural Safety*, 2026. (To be submitted)

Conference Papers:

- 1) M.A Ameri Fard Nasrand, Y. Li, "Risk-Based Inspection and Excavation Scheduling for Cracked Pipeline Systems Considering Regulatory Safety Constraints", *IPC2026, Calgary, Canada*.
- 2) M.A Ameri Fard Nasrand, Y. Li, "Inspection Scheduling for Crack Repair of Pipes considering Reliability-based Constraints", *CSCE Conference 2026, Montreal, Canada*.
- 3) M.A. Ameri Fard Nasrand, Peng H, Qiu W., Li Y, Adeeb S., "A Predictive Methodology for Estimating Runtime and Computational Resources in Complex Structural Reliability Problems Using Parallelized Monte Carlo Sampling", *CSCE Conference 2025, Winnipeg, Canada*.
- 4) M.A Ameri Fard Nasrand, S. Koduru, Y. Li, "Optimizing Crack Repair Policies using Value of Information Analysis considering Imperfect Inspection", *ICASP14, Dublin, Ireland, 2023*.
hdl.handle.net/2262/103634
- 5) M.A Ameri Fard Nasrand, Y. Li, "Calibrating Crack Depth Threshold for Pipeline Repair Policies", *CSCE Conference 2023, Moncton, Canada*.
doi.org/10.1007/978-3-031-61539-9_18
- 6) S. Koduru, M.A Ameri Fard Nasrand, Y. Li, "Sensitivity of Optimal Decision to Consequence Assumptions in VOI Analysis", *IFIP WG 7.5 Conference, Kyoto, Japan, 2022*.
doi.org/10.14989/ifipwg75_2022_4

AWARDS AND HONORS

Alberta Innovate Graduate Student Scholarship (2024).
Honor Award in Construction Innovation Center (CIC) Forum at the U of A (2024).
Third Place Award in CIC Forum at the U of A (2022).
Silver Medal in National Civil Engineering Olympiad in Iran (2018).
Ranked 5 among more than 30000 participants in Iranian MS.c. entrance exam (2018).
Scholarship Foundation Award from Iran's National Elite Foundation(2018).
Acceptance for M.Sc. degree as a talented student at Sharif University (2018).
Selected sportsman student in the Student Recognition Ceremony at SUT (2017).
Ranked 300 among 125000 participants in Iranian BS.c. entrance exam (2014).

REFERENCES

Prof. Yong Li <i>Associate Professor, University of Alberta</i>	yong9@ualberta.ca <i>PhD Supervisor</i>
Prof. Samer Adeeb <i>Professor, University of Alberta</i>	adeeb@ualberta.ca <i>PhD Supervisor Committee</i>
Prof. Mojtaba Mahsuli <i>Associate Professor, Sharif University of Technology</i>	mahsuli@sharif.edu <i>M.Sc. Supervisor</i>